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## Ranges Design Cleanup

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# 1 Abstract

[intro]

This paper proposes several small, independent design tweaks to Ranges that came up during LWG review of P0896 “The One Ranges Proposal” ([2]).

All wording sections herein are relative to the post-San Diego working draft.

## 1.1 Revision History

[intro.history]

### 1.1.1 Revision 1

[intro.history.r1]

- Rebase wording onto post-San Diego working draft.
- Strike section that suggested making the exposition-only concepts in [special.mem.concepts] available to users; this part of the proposal did not have consensus in LEWG.
- Update `safe_subrange_t` to account for potential dangling as well.

### 1.1.2 Revision 0

[intro.history.r0]

- In the beginning, all was *cv-void*. Suddenly, a proposal emerged from the darkness!

# 2 Deprecate `move_iterator::operator->` [disarm]

C++17 [iterator.requirements.general]/1 states:

... An iterator `i` for which the expression `(*i).m` is well-defined supports the expression `i->m` with the same semantics as `(*i).m`. ...

Input iterators are required to support the `->` operator ([input.iterators]), and `move_iterator` is an input iterator, so `move_iterator`’s arrow operator must satisfy that requirement, right? Sadly, it does not.

For a `move_iterator`, `*i` is an xvalue, so `(*i).m` is also an xvalue. `i->m`, however, is an lvalue. Consequently, `(*i).m` and `i->m` can produce observably different behaviors as subexpressions - they are not substitutable, as would be expected from a strict reading of “with the same semantics.” The fact that `->` cannot be implemented with “the same semantics” for iterators whose reference type is an rvalue was the primary motivation for removing the `->` requirement from the Ranges iterator concepts. It would benefit users to deprecate `move_iterator`’s `operator->` in C++20 as an indication that its semantics are *not* equivalent and that it will ideally go away some day.

## 2.1 Technical Specifications

[disarm.words]

- Strike `move_iterator::operator->` from the class template synopsis in [move.iterator]:

```
namespace std {
    template<class Iterator>
    class move_iterator {
        [...]
        constexpr iterator_type base() const;
        constexpr reference operator*() const;
        constexpr pointer operator->() const;

        constexpr move_iterator& operator++();
        constexpr auto operator++(int);
        [...]
    };
}
```

— Relocate the detailed specification of `move_iterator::operator->` from `[move.iter.elem]`:

```
constexpr reference operator*() const;
1      Effects: Equivalent to: return ranges::iter_move(current);

constexpr pointer operator->() const;
2      Returns: current.
```

```
constexpr reference operator[](difference_type n) const;
3      Effects: Equivalent to: ranges::iter_move(current + n);
```

to a new subclause “Deprecated `move_iterator` access” in Annex D between `[depr.iterator.primitives]` and `[depr.util.smartptr.shared.atomic]`:

```
1      The following member is declared in addition to those members specified in [move.iterator.elem]:
      namespace std {
          template<class Iterator>
          class move_iterator {
          public:
              constexpr pointer operator->() const;
          };
      }

      constexpr pointer operator->() const;
2      Returns: current.
```

## 3 *ref-view* ==> `ref_view` [ref]

The authors of P0896 added the exposition-only view type *ref-view* (P0896R4 `[range.view.ref]`) to serve as the return type of `view::all` (`[range.adaptors.all]`) when passed an lvalue container. *ref-view*<T> is an “identity view adaptor” – an adaptor which produces a view containing all the elements of the underlying range exactly – of a `Range` of type T whose representation consists of a T\*. A *ref-view* delegates all operations through that pointer to the underlying `Range`.

The LEWG-approved design from P0789R3 “Range Adaptors and Utilities” ([1]) used `subrange<iterator_t<R>, sentinel_t<R>>` as the return type of `view::all(c)` for an lvalue c of type R. *ref-view* and `subrange` are both identity view adaptors, so this change has little to no impact on the existing design. Why bother then? Despite that replacing `subrange` with *ref-view* in this case falls under as-if, *ref-view* has some advantages.

Firstly, a smaller representation: *ref-view* is a single pointer, whereas `subrange` is an iterator plus a sentinel, and sometimes a size. View compositions store many views produced by `view::all`, and many of those are views of lvalue containers in typical usage.

Second, and more significantly, *ref-view* is future-proof. *ref-view* retains the exact type of the underlying `Range`, whereas `subrange` erases down to the `Range`’s iterator and sentinel type. *ref-view* can therefore easily model any and all concepts that the underlying range models simply by implementing any required expressions via delegating to the actual underlying range, but `subrange` must store somewhere in its representation any properties of the underlying range needed to model a concept which it cannot retrieve from an iterator and sentinel. For example, `subrange` must store a size to model `SizedRange` when the underlying range is sized but does not have an iterator and sentinel that model `SizedSentinel`. If we discover in the future that it is desirable to have the `View` returned by `view::all(container)` model additional concepts, we will likely be blocked by ABI concerns with `subrange` whereas *ref-view* can simply add more member functions and leave its representation unchanged.

We’ve already realized these advantages for view composition by adding *ref-view* as an exposition-only `View` type returned by `view::all`, but users may like to use it as well as a sort of “Ranges `reference_wrapper`”.

### 3.1 Technical Specifications [ref.words]

— Update references to the name *ref-view* to `ref_view` in `[range.adaptors.all]`/2:

- 2 The name `view::all` denotes a range adaptor object (`[range.adaptor.object]`). For some  
 2 subexpression `E`, the expression `view::all(E)` is expression-equivalent to:
- 2 — `DECAY_COPY(E)` if the decayed type of `E` models `View`.
  - 2 — Otherwise, `ref_view{E}` if that expression is well-formed, ~~where `ref_view`~~  
~~is the exposition-only `View` specified below.~~
  - 2 — Otherwise, `subrange{E}`.
- (2.1) — Change the stable name `[range.view.ref]` to `[range.ref.view]` (for consistency with the stable names of  
 the other view classes defined in `[range]`), retitle to “class template `ref_view`” and modify as follows:

```
1 ref_view is a View of the elements of some other Range.
  namespace std::ranges {
    template<Range R>
      requires is_object_v<R>
      class ref_viewref_view : public view_interface<ref_viewref_view<R>> {
      private:
        R* r_ = nullptr; // exposition only
      public:
        constexpr ref_viewref_view() noexcept = default;

        template<not-same-as<ref_viewref_view> T>
          requires see below
          constexpr ref_viewref_view(T&& t);

        constexpr R& base() const;

        constexpr iterator_t<R> begin() const { return ranges::begin(*r_); }
        constexpr sentinel_t<R> end() const { return ranges::end(*r_); }

        constexpr bool empty() const
          requires requires { ranges::empty(*r_); }
          { return ranges::empty(*r_); }

        constexpr auto size() const requires SizedRange<R>
          { return ranges::size(*r_); }

        constexpr auto data() const requires ContiguousRange<R>
          { return ranges::data(*r_); }

        friend constexpr iterator_t<R> begin(ref_viewref_view r)
          { return r.begin(); }

        friend constexpr sentinel_t<R> end(ref_viewref_view r)
          { return r.end(); }
      };
  }

  template<not-same-as<ref_viewref_view> T>
    requires see below
    constexpr ref_viewref_view(T&& t);
    [...]
```

## 4 Comparison function object untemplates [untemp]

During LWG review of P0896’s comparison function objects (P0896R3 [range.comparisons]) we were asked, “Why are we propagating the design of the `std` comparison function objects, i.e. class templates that you shouldn’t specialize because they cannot be specialized consistently with the `void` specializations that you actually should be using?” For the Ranges TS, it was a design goal to minimize differences between `std`

and **ranges** to ease transition and experimentation. For the Standard, our goal should not be to minimize differences but to produce the best design. (As was evidenced by the LEWG poll in Rapperswil suggesting that we should not be afraid to diverge **std** and **ranges** components when there are reasons to do so.)

Absent a good reason to mimic the **std** comparison function objects exactly, we propose un-**template**-ing the **std::ranges** comparison function objects, leaving only concrete classes with the same behavior as the prior **void** specializations.

## 4.1 Technical specifications

[untemp.words]

In [functional.syn], modify the declarations of the **ranges** comparison function objects as follows:

[...]

```
namespace ranges {
    // [range.comparisons], comparisons
    template<class T = void>
    requires-see-below
    struct equal_to;

    template<class T = void>
    requires-see-below
    struct not_equal_to;

    template<class T = void>
    requires-see-below
    struct greater;

    template<class T = void>
    requires-see-below
    struct less;

    template<class T = void>
    requires-see-below
    struct greater_equal;

    template<class T = void>
    requires-see-below
    struct less_equal;

    template<> struct equal_to<void>;
    template<> struct not_equal_to<void>;
    template<> struct greater<void>;
    template<> struct less<void>;
    template<> struct greater_equal<void>;
    template<> struct less_equal<void>;
}
```

[...]

Update the specifications in [range.comparisons] as well:

- 2 There is an implementation-defined strict total ordering over all pointer values of a given type. This total ordering is consistent with the partial order imposed by the builtin operators **<**, **>**, **<=**, and **>=**.

```
template<class T = void>
    requires EqualityComparable<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, ==, const T&)
    struct equal_to {
        constexpr bool operator()(const T& x, const T& y) const;
    };
};
```

- 3 **operator()** has effects equivalent to: **return ranges::equal\_to<>{}(x, y);**

```
template<class T = void>
    requires EqualityComparable<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, ==, const T&)
    struct not_equal_to {
        constexpr bool operator()(const T& x, const T& y) const;
    };
};
```

```

};

4     operator() has effects equivalent to: return !ranges::equal_to<>{}(x, y);

template<class T = void>
    requires StrictTotallyOrdered<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, <, const T&)
struct greater {
    constexpr bool operator()(const T& x, const T& y) const;
};

5     operator() has effects equivalent to: return ranges::less<>{}(y, x);

template<class T = void>
    requires StrictTotallyOrdered<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, <, const T&)
struct less {
    constexpr bool operator()(const T& x, const T& y) const;
};

6     operator() has effects equivalent to: return ranges::less<>{}(x, y);

template<class T = void>
    requires StrictTotallyOrdered<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, <, const T&)
struct greater_equal {
    constexpr bool operator()(const T& x, const T& y) const;
};

7     operator() has effects equivalent to: return !ranges::less<>{}(x, y);

template<class T = void>
    requires StrictTotallyOrdered<T> || Same<T, void> || BUILTIN_PTR_CMP(const T&, <, const T&)
struct less_equal {
    constexpr bool operator()(const T& x, const T& y) const;
};

8     operator() has effects equivalent to: return !ranges::less<>{}(y, x);

template<> struct equal_to<void> {
    template<class T, class U>
        requires EqualityComparableWith<T, U> || BUILTIN_PTR_CMP(T, ==, U)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

9     Expects: If the expression std::forward<T>(t) == std::forward<U>(u) results in a call to a built-in
operator == comparing pointers of type P, the conversion sequences from both T and U to P shall be
equality-preserving ([concepts.equality]).

10    Effects:

(10.1) — If the expression std::forward<T>(t) == std::forward<U>(u) results in a call to a built-in
operator == comparing pointers of type P: returns false if either (the converted value of) t
precedes u or u precedes t in the implementation-defined strict total order over pointers of type P
and otherwise true.

(10.2) — Otherwise, equivalent to: return std::forward<T>(t) == std::forward<U>(u);

template<> struct not_equal_to<void> {
    template<class T, class U>
        requires EqualityComparableWith<T, U> || BUILTIN_PTR_CMP(T, ==, U)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

11    operator() has effects equivalent to:
        return !ranges::equal_to<<>{}(std::forward<T>(t), std::forward<U>(u));

```

```

template<> struct greater<void> {
    template<class T, class U>
        requires StrictTotallyOrderedWith<T, U> || BUILTIN_PTR_CMP(U, <, T)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

```

12      `operator()` has effects equivalent to:

```

    return ranges::less<>{}(std::forward<U>(u), std::forward<T>(t));

```

```

template<> struct less<void> {
    template<class T, class U>
        requires StrictTotallyOrderedWith<T, U> || BUILTIN_PTR_CMP(T, <, U)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

```

13      *Expects:* If the expression `std::forward<T>(t) < std::forward<U>(u)` results in a call to a built-in operator `<` comparing pointers of type `P`, the conversion sequences from both `T` and `U` to `P` shall be equality-preserving ([concepts.equality]). For any expressions `ET` and `EU` such that `decltype((ET))` is `T` and `decltype((EU))` is `U`, exactly one of `ranges::less<>{}(ET, EU)`, `ranges::less<>{}(EU, ET)`, or `ranges::equal_to<>{}(ET, EU)` shall be true.

14      *Effects:*

(14.1)      — If the expression `std::forward<T>(t) < std::forward<U>(u)` results in a call to a built-in operator `<` comparing pointers of type `P`: returns `true` if (the converted value of) `t` precedes `u` in the implementation-defined strict total order over pointers of type `P` and otherwise `false`.

(14.2)      — Otherwise, equivalent to: `return std::forward<T>(t) < std::forward<U>(u);`

```

template<> struct greater_equal<void> {
    template<class T, class U>
        requires StrictTotallyOrderedWith<T, U> || BUILTIN_PTR_CMP(T, <, U)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

```

15      `operator()` has effects equivalent to:

```

    return !ranges::less<>{}(std::forward<T>(t), std::forward<U>(u));

```

```

template<> struct less_equal<void> {
    template<class T, class U>
        requires StrictTotallyOrderedWith<T, U> || BUILTIN_PTR_CMP(U, <, T)
        constexpr bool operator()(T&& t, U&& u) const;

    using is_transparent = unspecified;
};

```

16      `operator()` has effects equivalent to:

```

    return !ranges::less<>{}(std::forward<U>(u), std::forward<T>(t));

```

Strip `<>` from occurrences of `ranges::equal_to<>`, `ranges::less<>`, etc. in: [defns.projection], [iterator.synopsis], [commonalgoreq.general]/2, [commonalgoreq.mergeable], [commonalgoreq.sortable], [range.syn], [range.adaptors.split\_view], [algorithm.syn], [alg.find], [alg.find.end], [alg.find.first.of], [alg.adjacent.find], [alg.count], [alg.mismatch], [alg.equal], [alg.is\_permutation], [alg.search], [alg.replace], [alg.remove], [alg.unique], [sort], [stable.sort], [partial.sort], [partial.sort.copy], [is.sorted], [alg.nth.element], [lower.bound], [upper.bound], [equal.range], [binary.search], [alg.merge], [includes], [set.union], [set.intersection], [set.difference], [set.symmetric.difference], [push.heap], [pop.heap], [make.heap], [sort.heap], [is.heap], [alg.min.max], [alg.lex.comparison],

and [alg.permutation.generators].

## 5 Reversing a reverse\_view [weiv\_\_esrever]

`view::reverse` in P0896 is a range adaptor that produces a `reverse_view` which presents the elements of the underlying range in reverse order - from back to front. `reverse_view` does so via the expedient mechanism of adapting the underlying view's iterators with `std::reverse_iterator`. Reversing a `reverse_view` produces a view of the elements of the original range in their original order. While this behavior is `correct`, it is likely to exhibit poor performance.

We propose that the effect of `view::reverse(r)` when `r` is an instance of `reverse_view` should be to simply return the underlying view directly. This behavior is both simple to specify and efficient to implement.

Similarly, reversing a `subrange` whose iterator and sentinel are `reverse_iterators` can be made more efficient by yielding a `subrange` of "unwrapped" iterators. Note that in this case we should take care to preserve any stored size information in the original subrange, since the size of the unwrapped base range is the same.

### 5.1 Technical specifications [sdrow.weiv\_\_esrever]

— Modify the specification of `view::reverse` in [range.reverse.adaptor] as follows:

- 1 The name `view::reverse` denotes a range adaptor object ([range.adaptor.object]). For some subexpression `E`, the expression `view::reverse(E)` is expression-equivalent to: `reverse_view{E}`.
- 1 — If the type of `E` is a cv-qualified specialization of `reverse_view`, `E.base()`.
- 1 — Otherwise, if the type of `E` is cv-qualified
- `subrange<reverse_iterator<I>, reverse_iterator<I>, K>`
- for some iterator type `I` and value `K` of `subrange_kind`,
- `subrange<I, I, K>(E.end().base(), E.begin().base(), E.size())`
- if `K` is `subrange_kind::sized` and
- `subrange<I, I, K>(E.end().base(), E.begin().base())`
- otherwise, except that in either case `E` is evaluated only once.
- 1 — Otherwise, `reverse_view{E}`.

## 6 Use cases left dangling [dangle]

What does this program fragment do in P0896?

```
std::vector<int> f();
o = std::ranges::copy(f(), o).out;
```

how about this one:

```
std::ranges::copy(f(), std::ostream_iterator<int>{std::cout});
```

The correct answer is, “These fragments are ill-formed because the iterator into the input range that `ranges::copy` returns would dangle - despite that the program fragment ignores that value - because LEWG asked us to remove the `dangling` wrapper and make such calls ill-formed.”

In the Ranges TS / revision one of P0896 an algorithm that returns an iterator into a range that was passed as an rvalue argument first wraps that iterator with the `dangling` wrapper template. A caller must retrieve the iterator value from the wrapper by calling a member function, opting in to potentially dangerous behavior explicitly. The use of `dangling` here makes it impossible for a user to inadvertently use an iterator that dangles.

In practice, the majority of range-v3 users in an extremely rigorous poll of the `#ranges` Slack channel (i.e., the author and two people who responded) never extract the value from a `dangling` wrapper. We prefer to

always pass lvalue ranges to algorithms when we plan to use the returned iterator, and use `dangling` only as a tool to help us avoid inadvertent use of potentially dangling iterators. Unfortunately, P0896 makes calls that would have used `dangling` in the TS design ill-formed which forces passing ranges as lvalues even when the dangling iterator value is not used.

We propose bringing back `dangling` in a limited capacity as a non-template tag type to be returned by calls that would otherwise return a dangling iterator value. This change makes the program fragments above well-formed, but without introducing the potentially unsafe behavior that LEWG found objectionable in the prior `dangling` design: there's no stored iterator value to retrieve.

## 6.1 Technical specifications

[dangle.words]

Modify the `<ranges>` synopsis in [ranges.syn] as follows:

```
namespace std::ranges {
    [...]

    // [range.range], Range
    template<class T>
        using iterator_t = decltype(ranges::begin(declval<T&>()));

    template<class T>
        using sentinel_t = decltype(ranges::end(declval<T&>()));

    template<forwarding-range R>
    using safe_iterator_t = iterator_t<R>;

    template<class T>
        concept Range = see below;

    [...]

    template<Iterator I, Sentinel<I> S = I, subrange_kind K = see below>
        requires (K == subrange_kind::sized || !SizedSentinel<S, I>)
        class subrange;

    // [dangling], dangling iterator handling
    struct dangling;

    template<Range R>
        using safe_iterator_t =
            conditional_t<forwarding-range<R>, iterator_t<R>, dangling>;

    template<forwarding-rangeRange R>
        using safe_subrange_t =
            conditional_t<forwarding-range<R>, subrange<iterator_t<R>>, dangling>;

    // [range.all], all view
    namespace view { inline constexpr unspecified all = unspecified; }

    [...]
}
```

Add a new subclause to [range.utility], following [range.subrange]:

### 23.6.4 Dangling iterator handling

[dangling]

1

The tag type `dangling` is used together with the template aliases `safe_iterator_t` and `safe_subrange_t` to indicate that an algorithm that typically returns an iterator into or subrange of a `Range` argument does not return an iterator or subrange which could potentially dangle for a particular rvalue `Range` argument which does not model `forwarding-range` ([range.range]).

```
namespace std {
    struct dangling {
        constexpr dangling() noexcept = default;
    };
}
```

```

    template<class... Args>
        constexpr dangling(Args&&...) noexcept { }
};
}

```

2

[*Example:*

```

vector<int> f();
auto result1 = ranges::find(f(), 42); // #1
static_assert(Same<decltype(result1), dangling>);
auto vec = f();
auto result2 = ranges::find(vec, 42); // #2
static_assert(Same<decltype(result2), vector<int>::iterator>);
auto result3 = ranges::find(subrange{vec}, 42); // #3
static_assert(Same<decltype(result3), vector<int>::iterator>);

```

The call to `ranges::find` at #1 returns `dangling` since `f()` is an rvalue `vector`; the `vector` could potentially be destroyed before a returned iterator is dereferenced. However, the calls at #2 and #3 both return iterators since the lvalue `vec` and specializations of `subrange` model forwarding-range. — *end example*]

## Bibliography

- [1] Eric Niebler. P0789r3: Range adaptors and utilities, 05 2018. <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p0789r3.pdf>.
- [2] Eric Niebler, Casey Carter, and Christopher Di Bella. P0896R4: The one ranges proposal, 11 2018. <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p0896r4.pdf>.